MEETING ROOM BOOKING SYSTEM

Software Requirement Specification

By

Mr. Thanarak	Chumsawat	68543210018-6
Mr. Tanapat	Chaitong	68543210064-0
Mr. Pathompon	Leejang	68543210034-3
Mr. Pasit	Petoumphun	68543210036-8
Mr. Wijitsin	Fangyen	68543210068-1

Software Engineering, Department of Electrical EngineeringRajamangala University of Technology Lanna

Project Advisor

_____

Mr. Sanya Uthayotha

Document Story

Version	Description	Date	Responsible	Status
Software Requirement Specification_V.0.1	Initial Draft - Add Chapter 1: Introduction - Add Chapter 2: Overall Description - Add Chapter 3: Functional Requirement	8 Jan 2026	TNR, PTP, PST	Draft
Software Requirement Specification_V.0.2	Update Requirement - Update URS based on feedback - Add ER Diagram (Proposed)	12 Jan 2026	JTS, TNP	Reviewed
Software Requirement Specification_V.1.0	First Release - Finalize SRS for Phase 1 approval	14 Jan 2026	SANTACLAUS TEAM	Released

Acronym
TNR = Thanarak Chumsawat(Product Owner)
TNP = Tanapat Chaitong (Scrum Master)
PTP = Pathompon Leejang (Developer)
PST = Pasit Petoumphun (Developer)
JTS = Wijitsin Fangyen (Developer)

Table of content

Chapter One | Introduction 1
1.1 Objective 1
1.2 Project Scope 1
1.3 Acronyms and Definitions 1
Chapter Two | Overall Description 2
2.1 Product Perspective 2
2.2 Product Features 2
2.3 User Classes and Characteristics 2
2.4 Operation Environment 3
Chapter Three | Functional Requiremen 4
3.1 User Requirement Specification 4
3.2 User Requirement Specification with the Software Requirement Specification 4
Chapter Four | Specific Requirement 6
4.1 Use Case Scenarios 6
4.2 Use Case Description 7
Chapter Five | References 18

Chapter One | Introduction

1.1 Objective

The objective of this Software Requirement Specification (SRS) is to define the requirements for the "Meeting Room Booking System" (Internal Organization Meeting Room Reservation System), covering both Web Application and Mobile Application platforms. The primary goal is to create a seamless and accurate meeting space management system that eliminates location conflicts and ensures smooth collaboration within the organization. The system aims to support booking for the entire organization with a 0% double booking rate, allowing users to complete a reservation within 4 clicks and providing accurate notifications.

This document serves as a guideline for developers, project managers, and stakeholders to ensure a unified understanding of the system design and development.

1.2 Project Scope

The Meeting Room Booking System consists of a Web Application and a Mobile Application designed to resolve current pain points such as double bookings, unknown room status, and lack of notifications . The scope of the system includes:

- **Feature #1: Real-time Availability & Booking:** Display real-time room status (Green/Red) and allow self-service booking and cancellation.
- **Feature #2: Conflict Detection:** An algorithm to prevent time overlapping and ensure Zero Double Booking.
- **Feature #3: Notification System:** Send email confirmations immediately after booking and reminders before the meeting starts.
- **Feature #4: Check-in & Auto-release:** A check-in system to confirm usage and automatically cancel bookings if the user does not show up (solving Ghost Booking).
- **Feature #5: Admin Management:** Tools for administrators to manage room information (add/edit/delete) and room status (e.g., closed for maintenance).
- **Feature #6: Approval Workflow:** An approval system for VIP rooms or large conference rooms.
- **Feature #7: Facility Reporting:** A system for users to report damaged equipment in meeting rooms.

1.3 Acronyms and Definitions

- **Double Booking:** A situation where a meeting room is booked by multiple parties for the same time slot.
- **Ghost Booking:** A situation where a room is booked but no one shows up, causing the room to appear occupied in the system.
- **PO (Product Owner):** The person responsible for defining the product vision and backlog.
- **SM (Scrum Master):** The person responsible for facilitating the team and removing obstacles.
- **Admin:** The system administrator or building staff who manages room data and approvals.
- **User:** General employees who use the system to book rooms.

Chapter Two | Overall Description

2.1 Product Perspective

The Meeting Room Booking System is developed to replace the existing manual booking process, which relies on walking to check rooms, calling staff, or using paper/Excel logs, leading to delays and errors .

This new system operates as a centralized digital platform connecting General Users, Administration/Operations, and the System itself . It enables Self-Service Booking, allowing users to check status and book rooms via Web or Mobile App without waiting for a middleman, while the system automatically handles logic and verification to reduce human error .

2.2 Product Features

Based on User Needs and the Product Backlog, the major features are categorized as follows:

- **Booking Management:** Search for available rooms, create bookings, and cancel bookings, supporting both ad-hoc and recurring reservations.
- **Access Control & Verification:** Verify conditions for VIP rooms and require Check-in to confirm presence.
- **Automated Alerts:** Automatically send email notifications for confirmations and reminders.
- **Administrative Tools:** Manage room details, approve requests, and handle facility reports.
- **History & Optimization:** View booking history and provide smart room recommendations based on capacity.

2.3 User Classes and Characteristics

The users of the system are categorized by their roles and behaviors:

1. **General Employees (e.g., Marketing, Sales, Interns):**
    - The largest user group. They need to book rooms quickly (Ad-hoc) and mostly use mobile phones. They require a user-friendly interface to reduce the learning curve.
2. **Planners & Secretaries (e.g., HR, Executive Secretary):**
    - Users who manage schedules for others or organize training. They need Recurring Booking features and the ability to edit or swap rooms easily and accurately.
3. **Approvers & Executives (e.g., Gatekeeper, CEO):**
    - Users with authority to approve large rooms or VIP rooms. Executives require privacy and priority booking.
4. **Support Staff (e.g., Admin, IT Support, Housekeeper):**
    - Staff responsible for room maintenance and cleanliness. They need to see room status to perform their duties or block rooms for repairs.

2.4 Operation Environment

The system is designed to operate in a specific environment to ensure stability and performance. The following hardware and software configurations are required for the development and deployment of the "Meeting Room Booking System".

Hardware Interface The application has been tested and optimized for the following workstation specifications (Client/Development Side):

- Apple MacBook Pro M4
- HP Victus, HP Victus 16 (s0095ax)
- Acer Nitro V 15 (ANV15-51-50UV)

Software Interface The system utilizes the following software technologies and infrastructure for backend operations and deployment:

- Cloud Infrastructure: AWS EC2 (Amazon Elastic Compute Cloud)
- Database & Backend as a Service: Supabase (Superbase)
- Containerization: Docker
- Framework/Library: Antigravity

Chapter Three | Functional Requirement

3.1 User Requirement Specification

Feature #1 Booking System
URS-01: Users can view the real-time availability of meeting rooms on the web/mobile application.
URS-02: Users can create a booking by inputting date, time, room, topic, and number of attendees.
URS-03: Users can insure that the booking will not overlap with others (Zero Double Booking).
URS-04: Users can cancel their own booking on the application.
URS-11: Users can get a room recommendation based on the number of attendees.
URS-12: Users can view their own booking history.

Feature #2 Notification & Check-in System
URS-05: Users receive email confirmation immediately after booking and reminders before the meeting starts.
URS-06: Users must check-in to confirm the usage of the room to prevent ghost booking.

Feature #3 Policy & Fairness
URS-07**:** Users are limited to recurring bookings for a maximum of 4 weeks to ensure fairness.

Feature #4 Admin & Management System
URS-08: Administrators can manage room information (add/edit/delete) and room status.
URS-09: Administrators can approve or reject the request for VIP rooms.
URS-10: Users can report damaged equipment in the meeting room.

3.2 User Requirement Specification with the Software Requirement Specification

URS-01: Users can view the real-time availability of meeting rooms on the web/mobile application.
SRS-001: The system shall display a dashboard showing a calendar view of all meeting rooms.
SRS-002: The system shall indicate the status of each room using color codes (e.g., Green for Available, Red for Occupied).
SRS-003: The system shall load the availability status within 2 seconds.
SRS-004: The system shall update the status in real-time without requiring a manual page refresh.

URS-02: Users can create a booking by inputting date, time, room, topic, and number of attendees.
SRS-005: The system shall provide a "Create Booking" form.
SRS-006: The system shall require the user to input the Meeting Topic, Date, Start Time, End Time, and Number of Attendees.
SRS-007: The system shall display an error message "Cannot book past time" if the user selects a date/time in the past.
SRS-008: The system shall generate a unique Booking ID upon successful booking.

URS-03: Users can insure that the booking will not overlap with others (Zero Double Booking).
SRS-009: The system shall validate the requested time slot against the database before saving.
SRS-010: The system shall display an error message "Room is not available" if the requested time overlaps with an existing booking record.
SRS-011: The system shall lock the record during the transaction to prevent concurrent booking conflicts.

URS-04: Users can cancel their own booking on the application.
SRS-012: The system shall provide a "Cancel" button for active bookings in the user's dashboard.
SRS-013: The system shall allow cancellation only if the logged-in user is the owner of the booking (except Admin).
SRS-014: The system shall immediately update the room status to "Available" (Green) upon cancellation.

URS-05: Users receive email confirmation immediately after booking and reminders before the meeting starts.
SRS-015: The system shall send an automated email to the user's registered email address upon "Confirmed" status.
SRS-016: The email content shall include Date, Time, Room Name, and Meeting Topic.
SRS-017: The system shall send a notification/email reminder 15 minutes before the meeting start time.

URS-06: Users must check-in to confirm the usage of the room to prevent ghost booking.
SRS-018: The system shall display a "Check-in" button or QR Code scanner 15 minutes before and after the start time.
SRS-019: The system shall automatically change the booking status to "Cancelled" if the user does not check-in within 15 minutes after the start time.
SRS-020: The system shall release the room to be "Available" for others immediately after an auto-cancellation.

URS-08: Administrators can manage room information (add/edit/delete) and room status.
SRS-021: The system shall provide an Admin Dashboard for room management.
SRS-022: The system shall allow the Admin to disable a room (set status to "Maintenance" or "Inactive") temporarily.
SRS-023: The system shall update the user-facing calendar immediately when an Admin adds or removes a room.

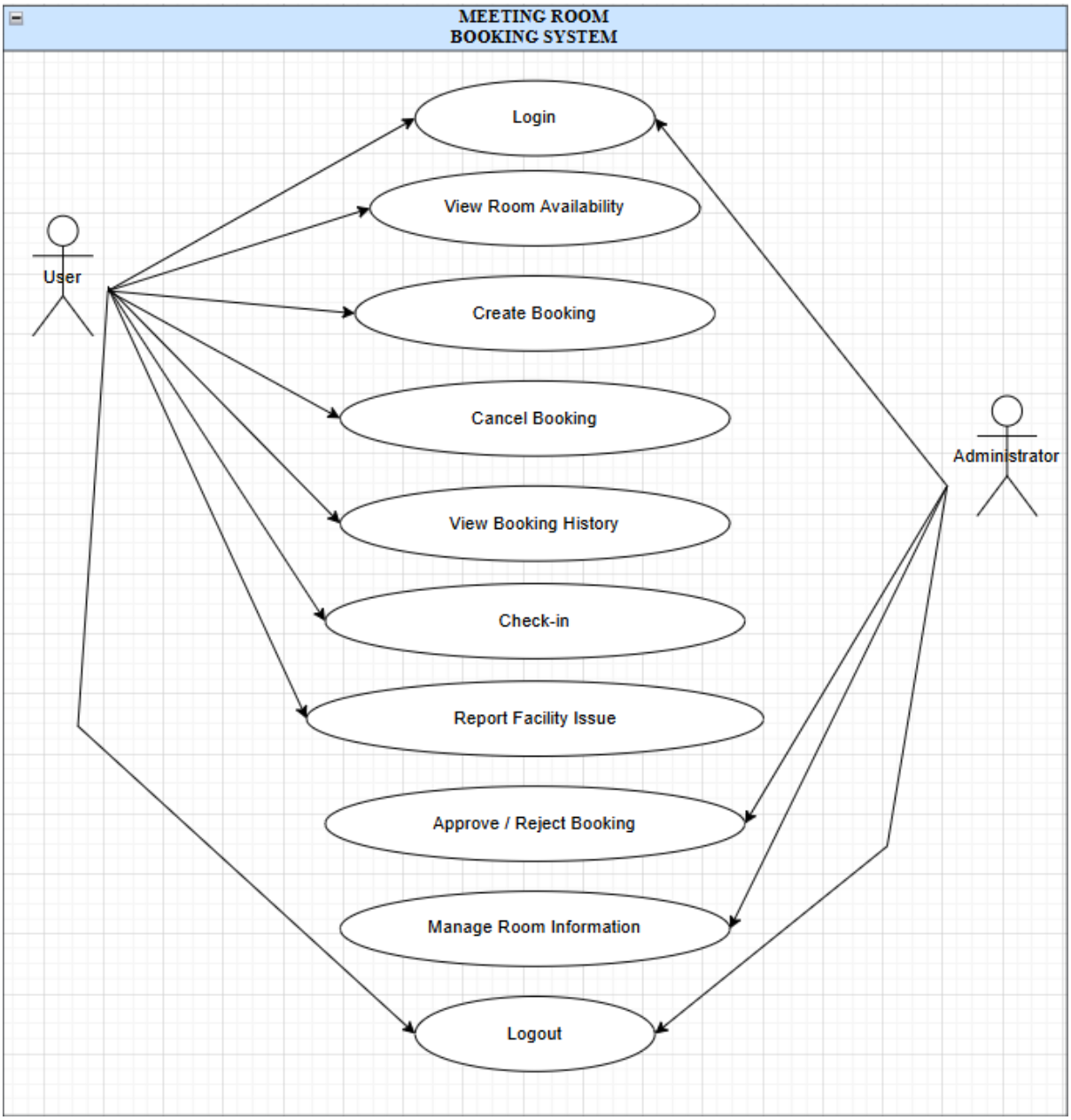
URS-09: Administrators can approve or reject the request for VIP rooms.
SRS-024: The system shall set the initial status of bookings for VIP rooms to "Pending Approval".
SRS-025: The system shall notify the Administrator when a Pending request is created.
SRS-026: The system shall provide "Approve" and "Reject" buttons for the Administrator.
SRS-027: The system shall change the status to "Confirmed" only after the Administrator clicks "Approve".

URS-10: Users can report damaged equipment in the meeting room.
SRS-028: The system shall provide a "Report Issue" form allowing users to select the equipment (e.g., Projector, Air Conditioner).
SRS-029: The system shall send a notification to the Admin or IT Support upon submission of a report.
SRS-030: The system shall allow the Admin to block the room for maintenance based on the report.

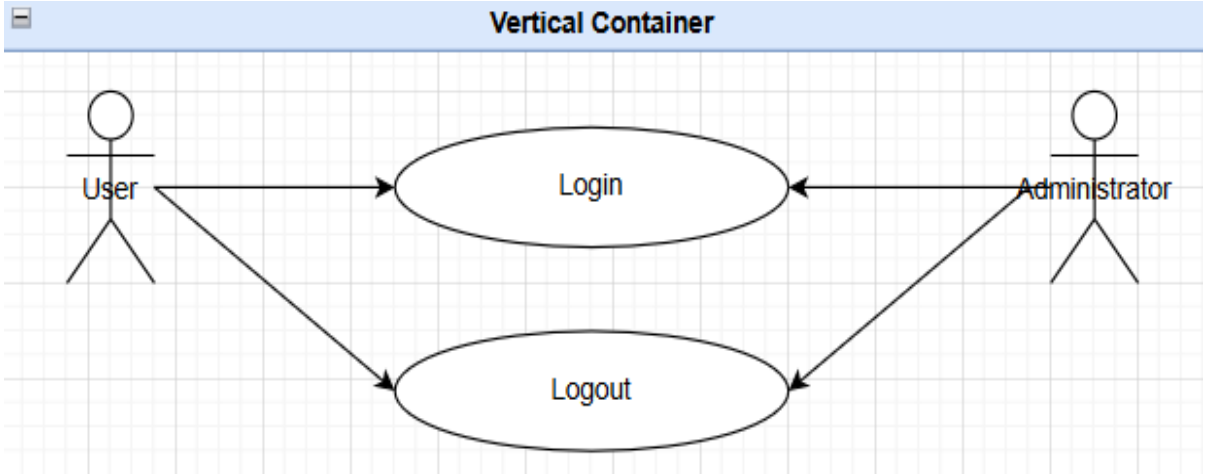
Chapter Four | Specific Requirement

4.1 Use Case Scenarios

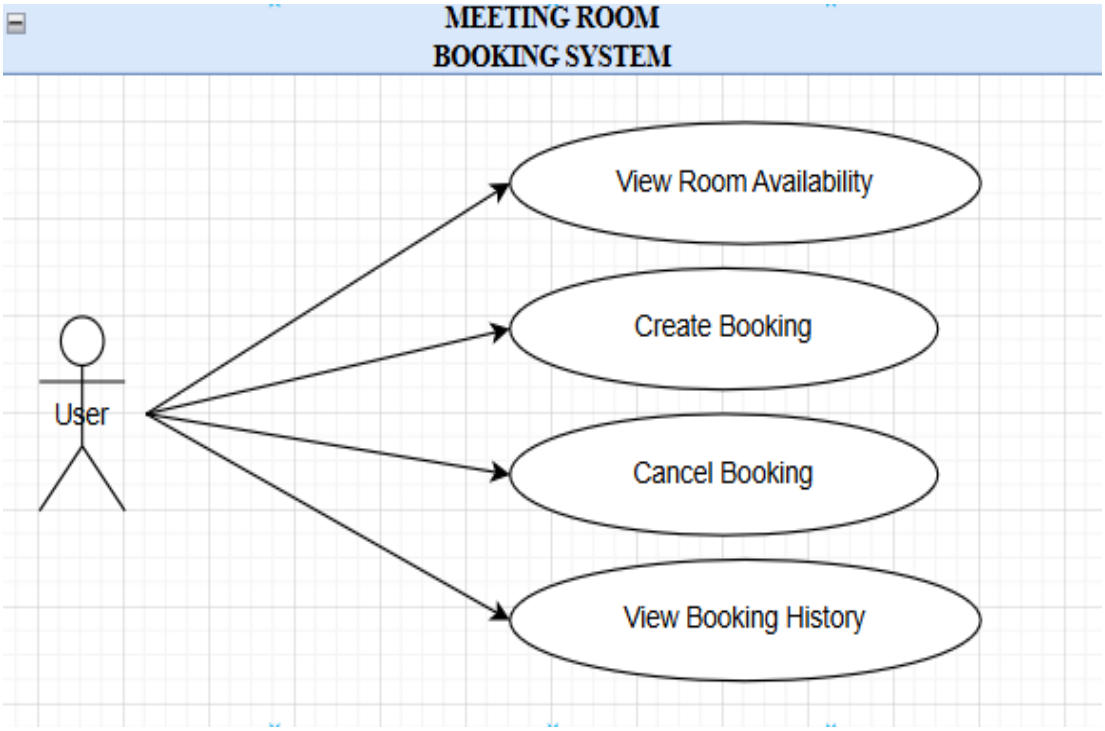
4.1.1 Use Case Diagram (Overall System)



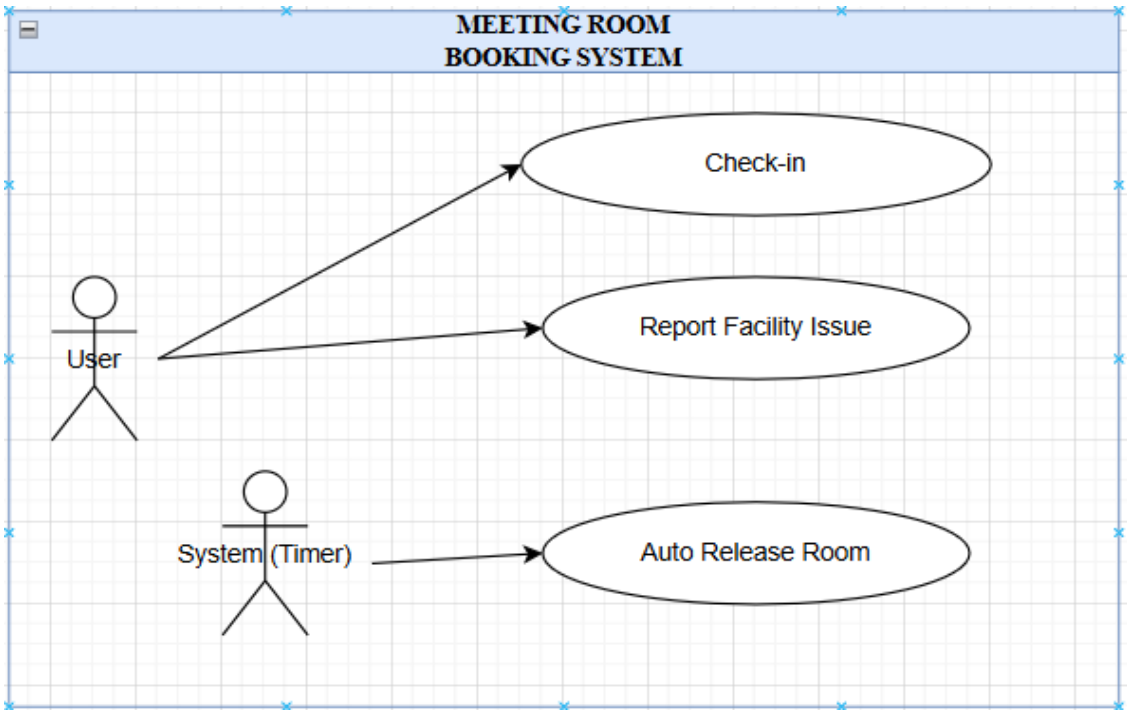
4.1.2 Use Case Diagram (Authentication System)



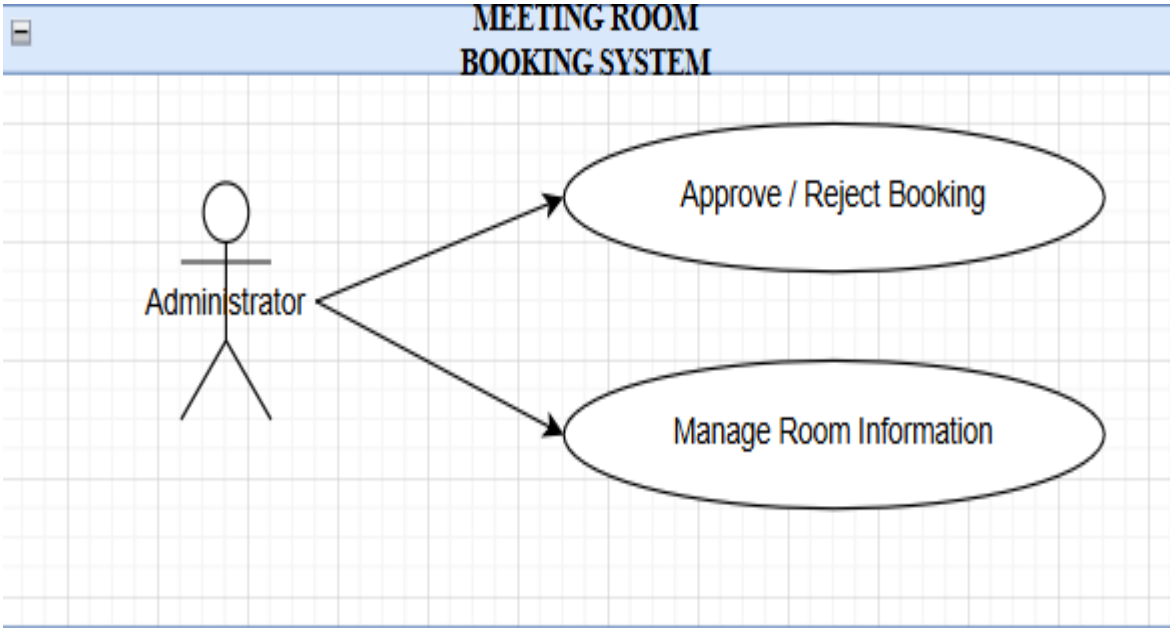
4.1.3 Use Case Diagram (Booking System)



4.1.4 Use Case Diagram (Action & Notification System)



4.1.5 Use Case Diagram (Management System)



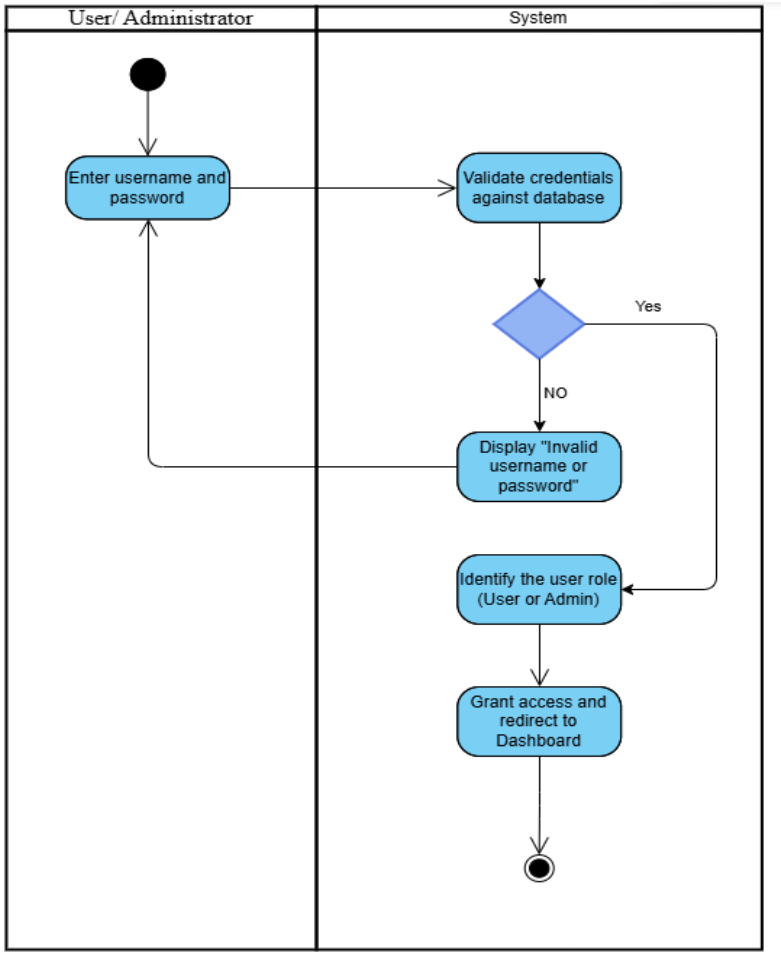
4.2 Use Case Description

4.2.1 Feature#1: Authentication System

UC-01: Users and Administrators can log in to the system to verify their identity and access specific features.

Topic	Description
Use Case Name	Login
Actor	User, Administrator
Description	Users log in to the system to access booking features and identify their roles.
Trigger	User opens the application or website.
Pre-conditions	User must have a registered account.
Post-condition	User is authenticated and redirected to the Dashboard.
Normal Flows	1. User enters username and password. 2. System validates credentials against the database. 3. System identifies the user role (User or Admin). 4. System grants access to the main dashboard.
Exceptions	**1. Invalid Credentials:** System displays "Invalid username or password" and asks the user to retry.

Activity Diagram :(AD-01):

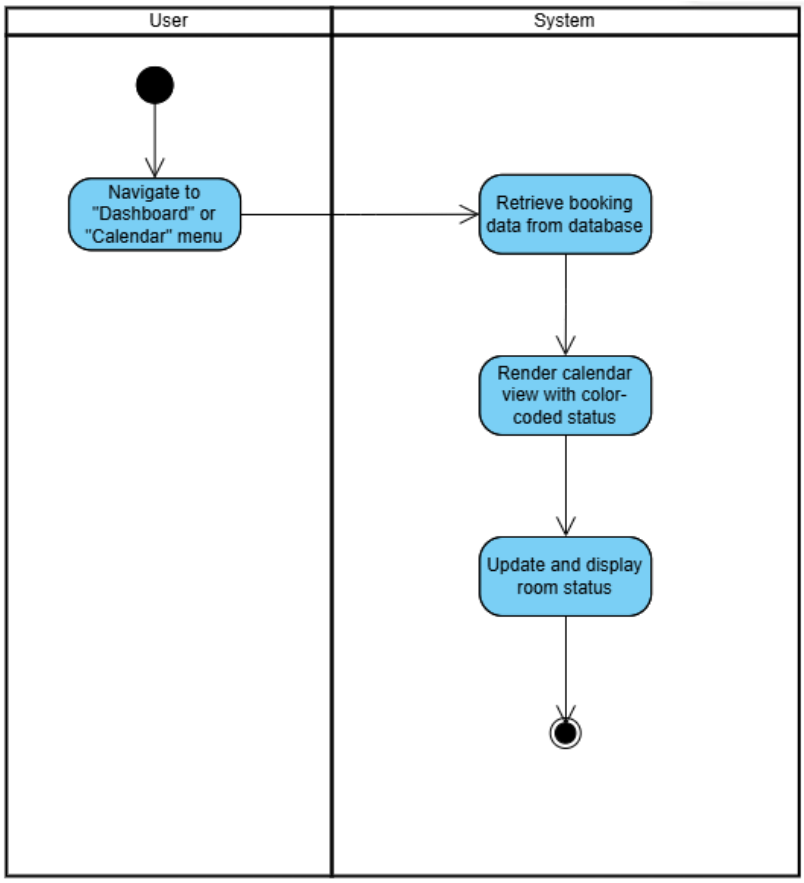


Feature #2: Booking System

UC-02: Users can view the real-time availability of meeting rooms on the dashboard.

Topic	Description
Use Case Name	View Room Availability
Actor	User
Description	Users view the real-time status of meeting rooms to find available slots.
Trigger	User navigates to the "Dashboard" or "Calendar" menu.
Pre-conditions	User is logged in.
Post-condition	System displays room status (Green=Available, Red=Occupied).
Normal Flows	1. System retrieves booking data from the database. 2. System renders the calendar view with color-coded status. 3. User filters by date, time, or room capacity.

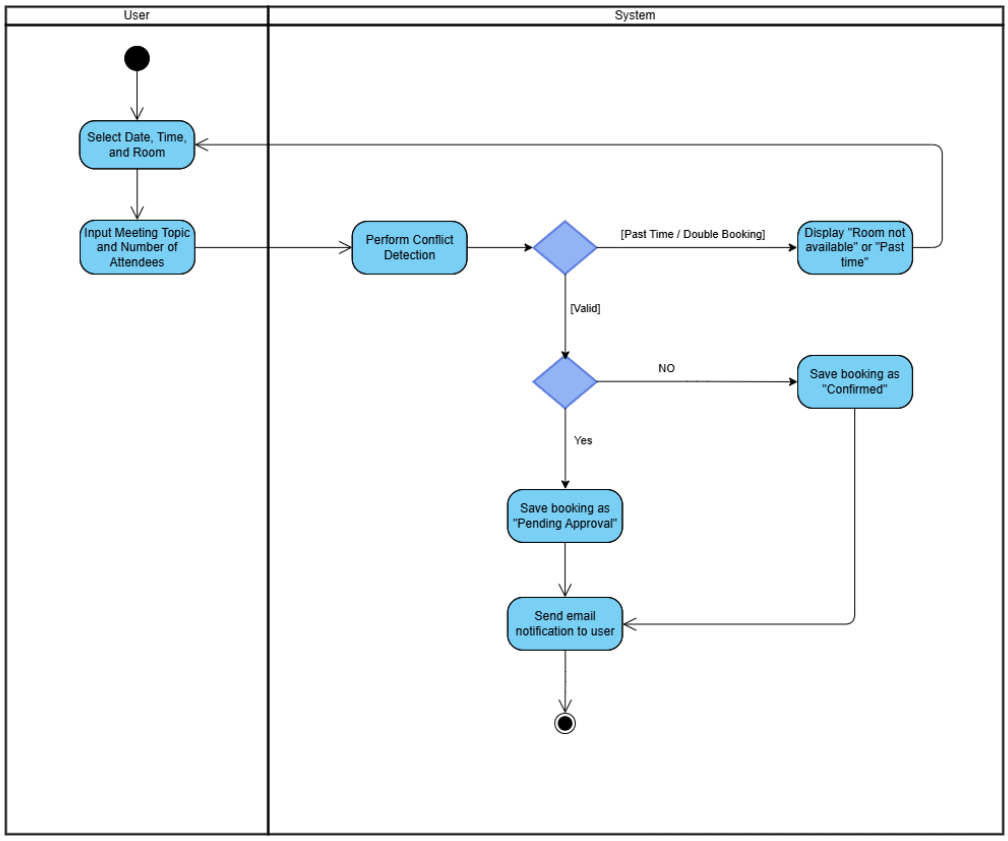
Activity Diagram :(AD-02):



UC-03: Users can create a new booking by specifying the date, time, room, and meeting topic.

Topic	Description
Use Case Name	Create Booking
Actor	User
Description	Users create a new reservation for a meeting room.
Trigger	User clicks "Book Now" or selects a time slot.
Pre-conditions	The selected room must be available.
Post-condition	Booking is saved, and a confirmation email is sent.
Normal Flows	1. User selects Date, Time, and Room. 2. User inputs Meeting Topic and Number of Attendees. 3. System performs **Conflict Detection** to ensure no overlap. 4. **If General Room:** System saves booking as "Confirmed". 5. **If VIP Room:** System saves booking as "Pending Approval". 6. System sends an email notification to the user.
Exceptions	**1. Double Booking:** If the slot is taken during the process, system alerts "Room not available". **2. Past Time:** System rejects bookings for past dates/times.

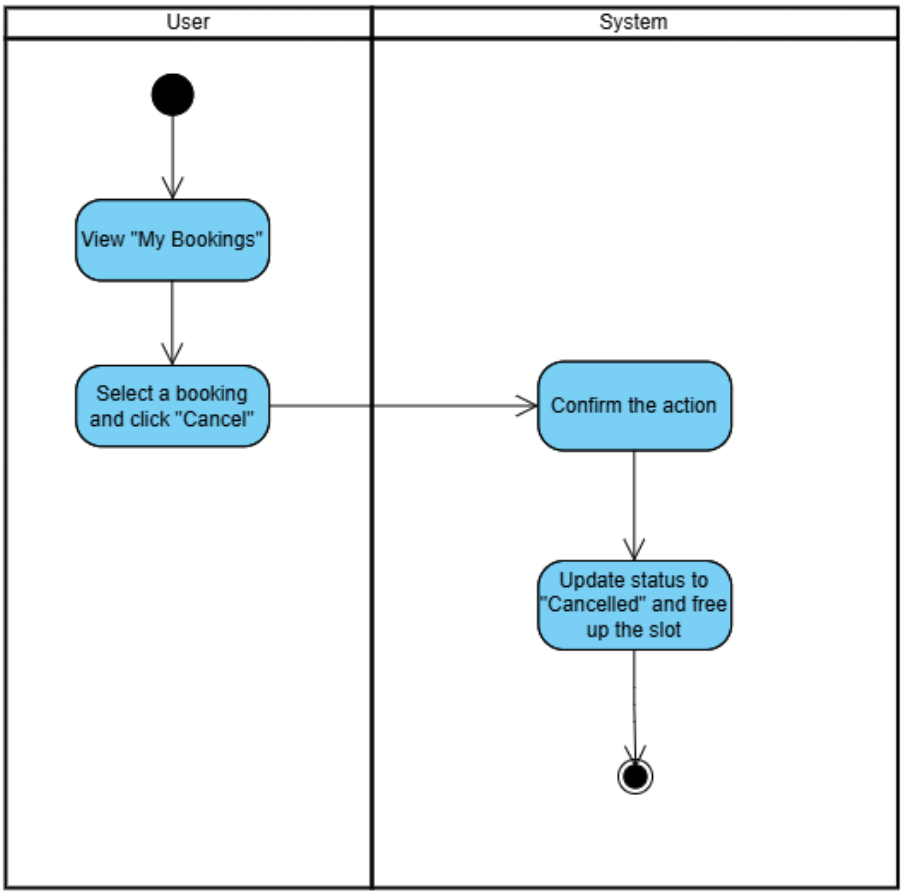
Activity Diagram :(AD-03):



UC-04: Users can cancel their own active bookings to release the room for others.

Topic	Description
Use Case Name	Cancel Booking
Actor	User (Owner)
Description	Users cancel their own existing bookings.
Trigger	User clicks "Cancel" on a specific booking item.
Pre-conditions	The booking must not have expired.
Post-condition	Room status becomes "Available".
Normal Flows	1. User views "My Bookings". 2. User selects a booking and clicks "Cancel". 3. System confirms the action. 4. System updates status to "Cancelled" and frees up the slot.

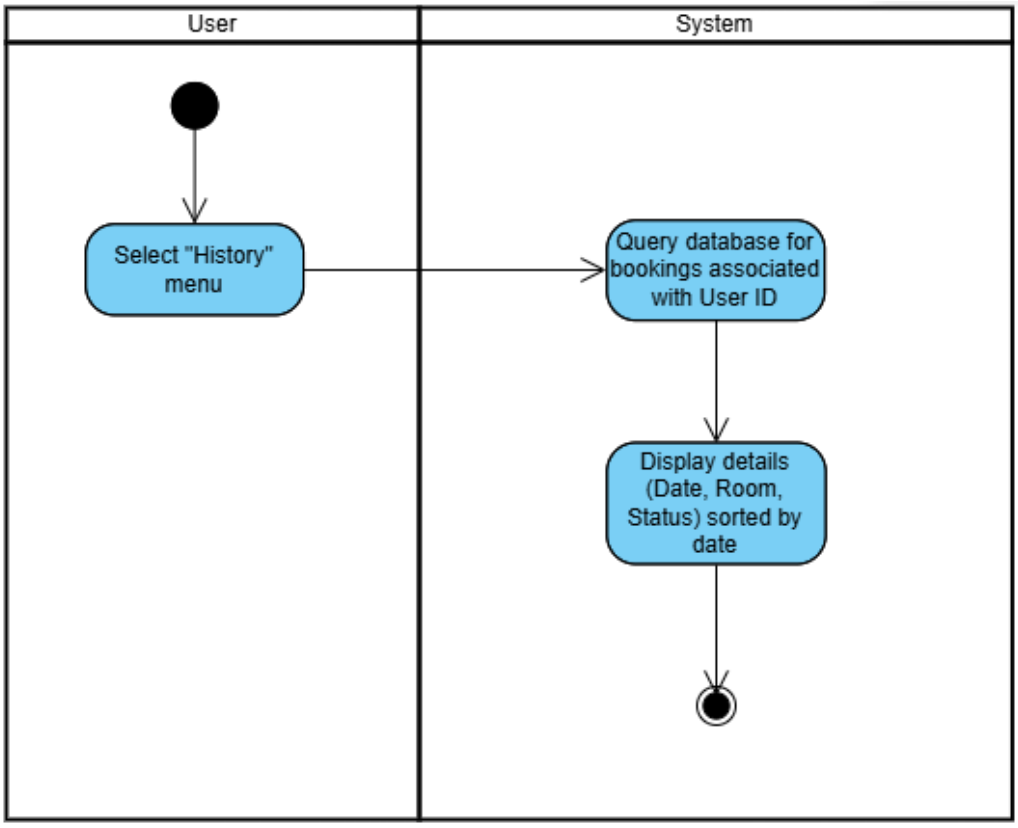
Activity Diagram :(AD-04):



UC-05: Users can view a history of their past and upcoming meeting room reservations.

Topic	Description
Use Case Name	View Booking History
Actor	User
Description	Users view a list of their past and upcoming bookings.
Trigger	User selects "History" menu.
Pre-conditions	User is logged in.
Post-condition	A list of bookings is displayed.
Normal Flows	1. System queries the database for bookings associated with the User ID. 2. System displays details (Date, Room, Status) sorted by date.

Activity Diagram :(AD-05):

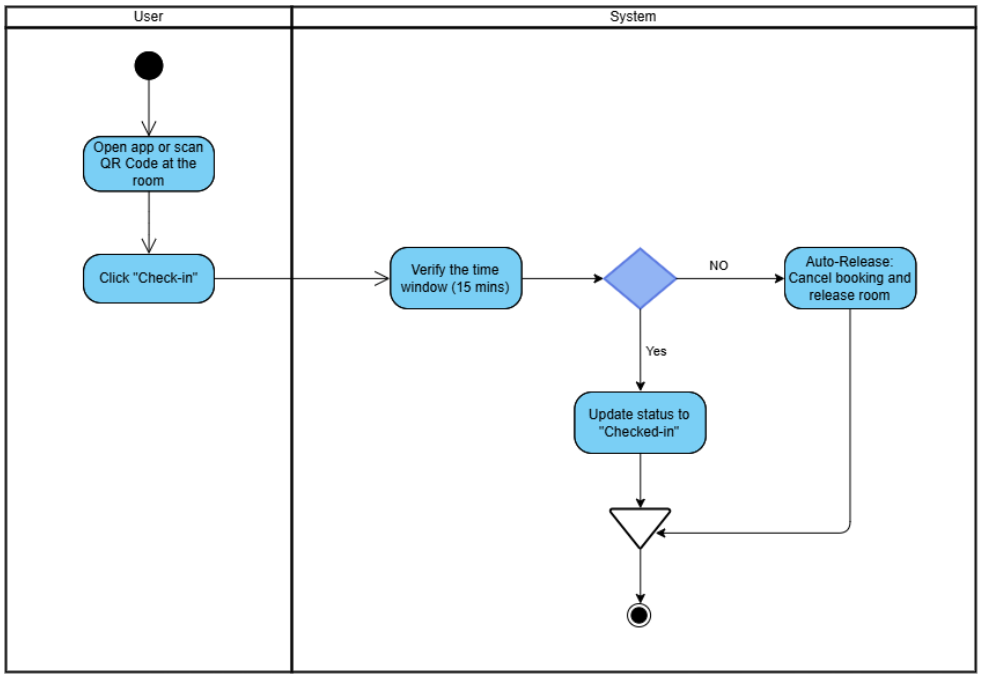


Feature #3: Action & Notification System

UC-06: Users can check-in at the meeting room to confirm their presence and prevent auto-cancellation.

Topic	Description
Use Case Name	Check-in
Actor	User
Description	Users confirm their presence to prevent "Ghost Booking".
Trigger	User arrives at the room within the specific timeframe (15 mins before/after start).
Pre-conditions	User has a valid booking for the current time.
Post-condition	Booking is marked as "Checked-in".
Normal Flows	1. User opens the app or scans QR Code at the room. 2. User clicks "Check-in". 3. System verifies the time window. 4. System updates status to "Checked-in".
Exceptions	**1. Auto-Release:** If no Check-in occurs within 15 minutes after start time, System automatically cancels the booking and releases the room.

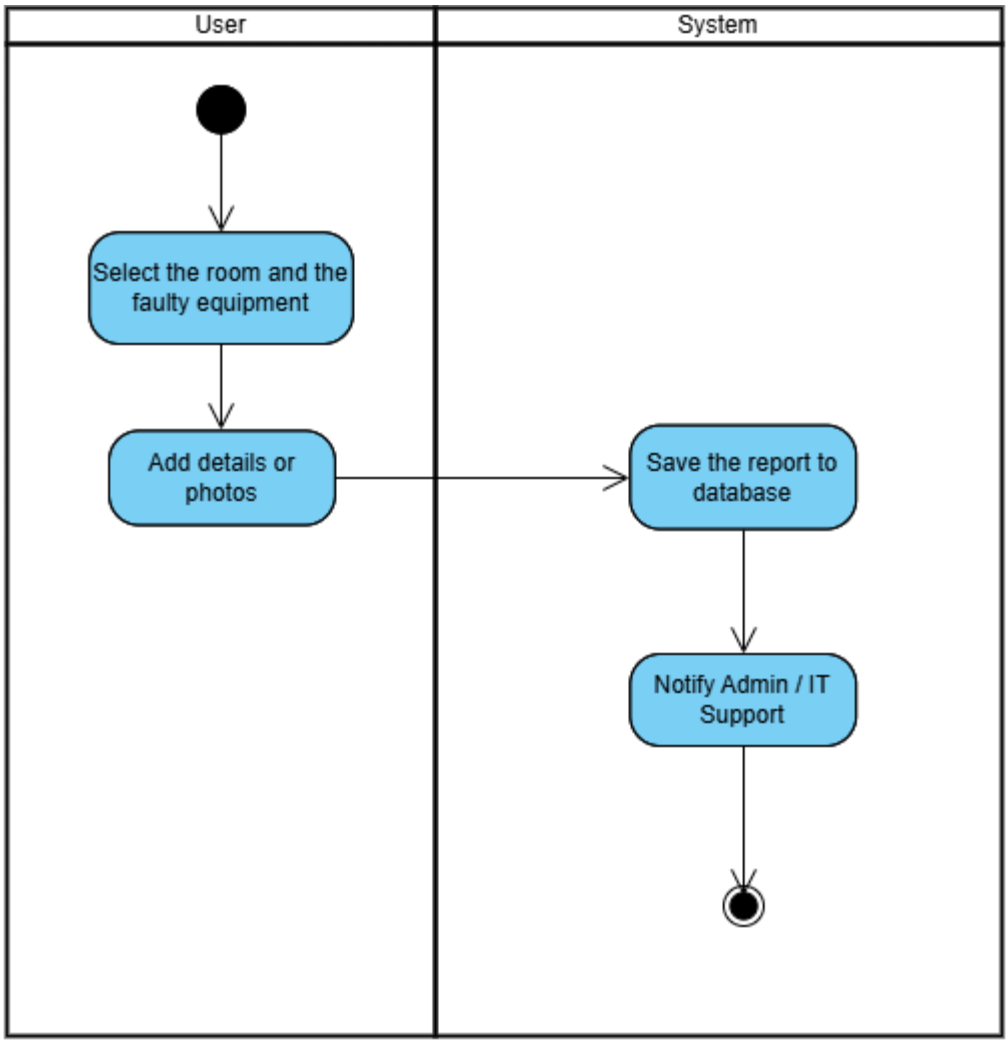
Activity Diagram :(AD-06):



UC-07: Users can report damaged equipment or facility issues to the administration.

Topic	Description
Use Case Name	Report Facility Issue
Actor	User
Description	Users report damaged equipment (e.g., broken projector, air con).
Trigger	User clicks "Report Issue".
Pre-conditions	User is logged in.
Post-condition	Admin receives a notification.
Normal Flows	1. User selects the room and the faulty equipment. 2. User adds details or photos. 3. System saves the report and notifies Admin/IT Support.

Activity Diagram :(AD-07):

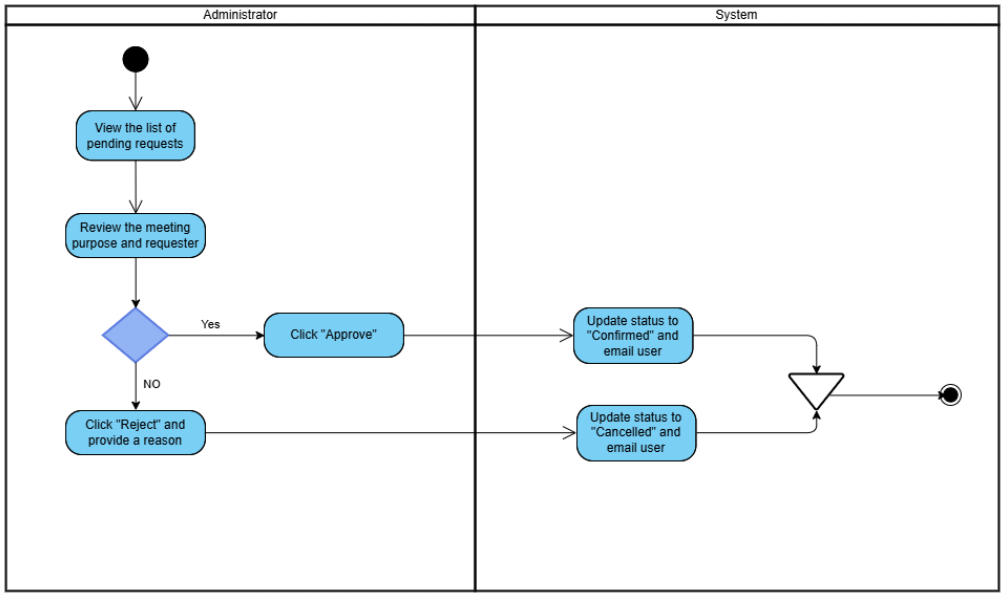


Feature #4: Management System

UC-08: Administrators can approve or reject booking requests for VIP meeting rooms.

Topic	Description
Use Case Name	Approve/Reject Booking
Actor	Administrator
Description	Admins review and decide on booking requests for VIP rooms.
Trigger	Admin receives a "Pending Approval" request.
Pre-conditions	A VIP room booking request exists.
Post-condition	Booking status changes to "Confirmed" or "Cancelled".
Normal Flows	1. Admin views the list of pending requests. 2. Admin reviews the meeting purpose and requester. 3. Admin clicks **"Approve"**. 4. System updates status to "Confirmed" and emails the user.
Alternate Flows	**3a. Reject:** Admin clicks "Reject" and provides a reason. System updates status to "Cancelled".

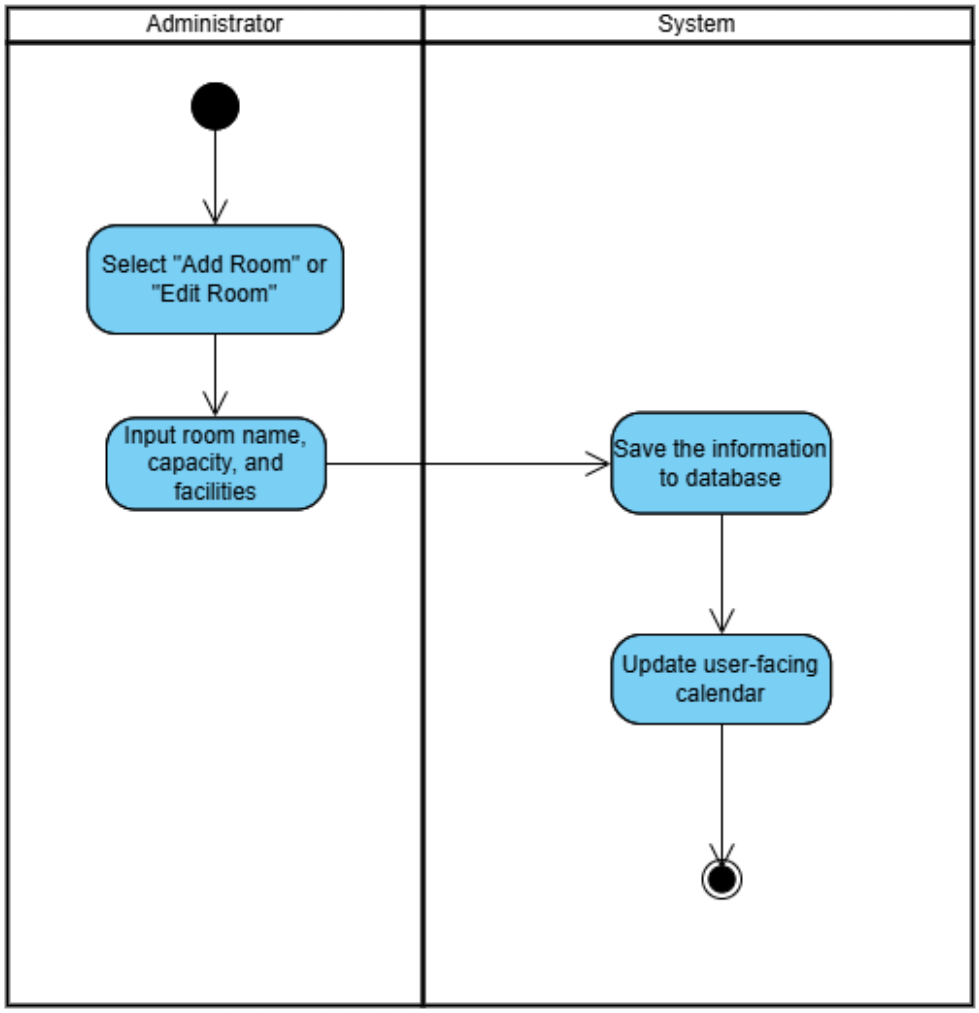
Activity Diagram :(AD-08):



UC-09: Administrators can add, edit, or disable meeting room information in the system.

Topic	Description
Use Case Name	Manage Room Information
Actor	Administrator
Description	Admins add, edit, or disable meeting rooms in the system.
Trigger	Admin accesses "Room Management".
Post-condition	Database is updated with new room details.
Normal Flows	1. Admin selects "Add Room" or "Edit Room". 2. Admin inputs room name, capacity, and facilities. 3. System saves the information. 4. New room info is immediately visible to users.

Activity Diagram :(AD-09):



Chapter Five | References

[1] Definition of User Interface.
http://searchsoa.techtarget.com/definition/user-interface

[2] Definition of Use Case Diagram.
https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-use-case-diagram/

[3] Definition of Activity Diagram.
https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-activity-diagram/